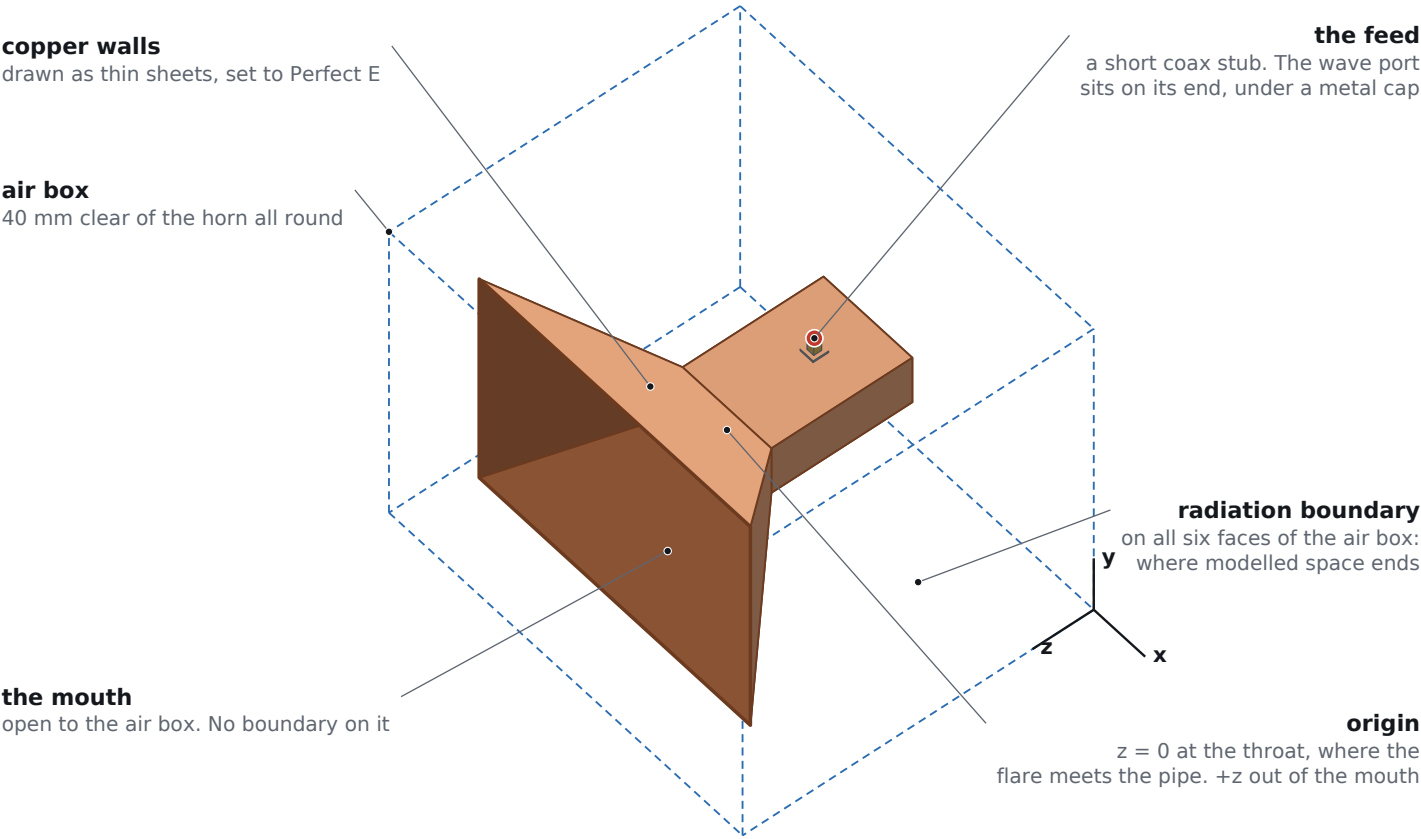


Designing the horn in Ansys HFSS

The basics of antenna design in HFSS, worked through on this project's horn: what the antenna has to do, how the solver works, how to build the model, and how to tell a good answer from a wrong one.



The model this guide builds: the horn as thin metal sheets, a coax feed on the top wall, and a box of air around it.

the numbers to beat	from theory	accept from HFSS
peak gain	13.4 dBi	11.9 to 14.9 dBi
upper limit for this mouth	16.3 dBi	never above it
beam width, E-plane	33.4°	29° to 37°
beam width, H-plane	35.2°	31° to 39°
S11 across 2.400–2.4835 GHz	set by the probe	–10 dB or lower, after tuning

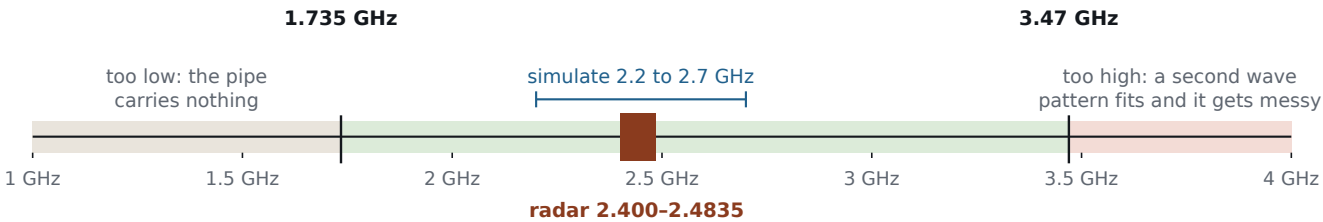
1 What the horn has to do

An antenna design comes down to three numbers. For this horn they are:

what it means		target
gain	how strongly the horn focuses power straight ahead, compared with an antenna that spreads it evenly in every direction. Measured in dBi	13.4 dBi, about 22 times
beam width	how wide the beam is, measured between the two angles where the power has dropped to half (−3 dB)	about 33° × 35°
match, S11	how much of the power sent up the cable bounces back off the horn instead of going out as the beam	−10 dB or lower across the band

How a horn works

The pipe at the back is a waveguide: a metal tube that carries a radio wave along it. A tube this size only carries signals above 1.735 GHz, and above 3.47 GHz a second wave pattern starts to fit and the signal gets messy. The radar's band sits comfortably between the two, so the pipe carries exactly one clean wave. The flare then widens that wave gently into a large mouth, and a large mouth makes a narrow, strong beam.



Gain has a ceiling

For a mouth of 263.8 × 193.1 mm at 2.45 GHz, no antenna can do better than 16.3 dBi. Real horns reach about half of that ceiling: integrating the wave across this horn's mouth gives 51%, which is 13.4 dBi. This is the most useful check in the whole guide. **If HFSS reports more than 16.3 dBi, the model is wrong**, however well it converged.

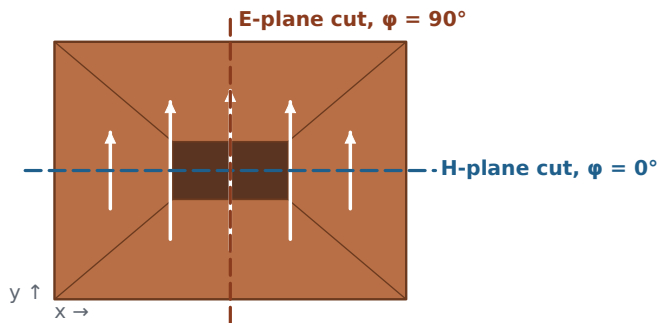
S11, in plain terms

S11 = −10 dB means:	S11	power reflected	power into the horn
100% sent up the cable	−3 dB	50 %	50 % (bad)
	−6 dB	25 %	75 %
90% goes into the horn and out as the beam 10% comes back	−10 dB	10 %	90 % (the target)
	−15 dB	3 %	97 %
	−20 dB	1 %	99 %

The flare sets the gain and the beam, and theory already gets those right. The match depends on the probe, and nothing but a simulation or a measurement tells you that. Getting the match right is the real reason to run HFSS.

The E-plane and the H-plane

A radiation pattern is 3D, so it is usually shown as two slices through the beam. They are named after the electric field inside the horn.



Looking into the mouth

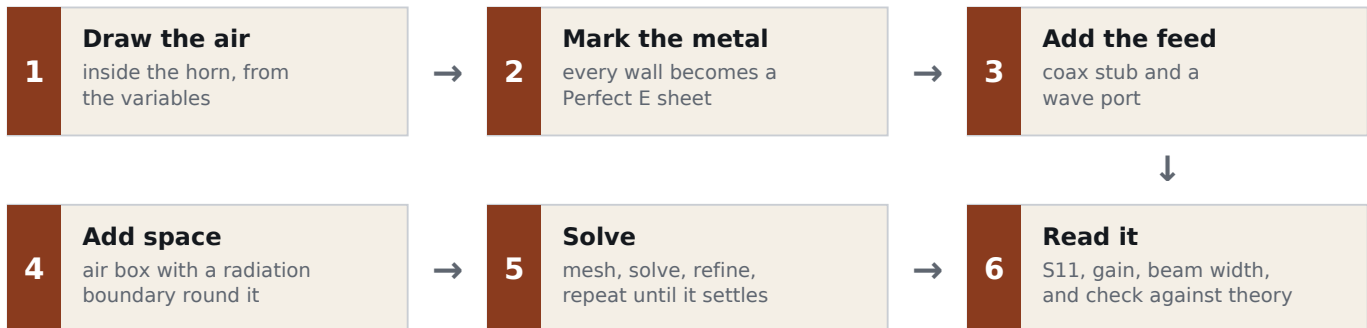
White arrows: the electric field. It points across the narrow side (along y), strongest in the middle and fading to nothing at the side walls.

The E-plane is the slice that contains those arrows. The H-plane is the slice at right angles to it.

Mixing the two up is the most common mistake in reporting a horn. There is an easy check: here **the E-plane beam is the narrower one** (33.4° against 35.2°).

2 How HFSS works

HFSS solves for the electric and magnetic field everywhere inside a box you define. You never write equations. You describe the geometry and say what each surface is: metal, the place the signal enters, or the edge of the modelled world.



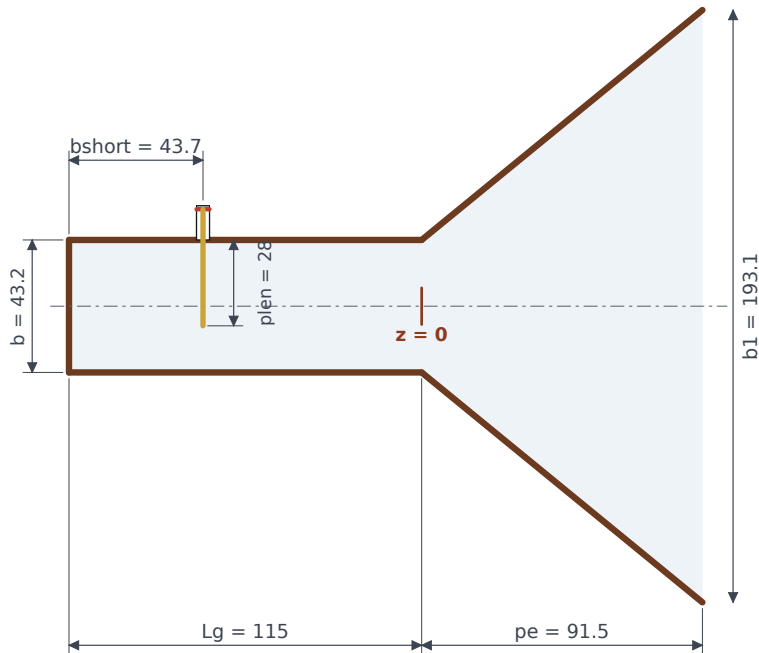
- **Mesh.** HFSS fills the space with small tetrahedra and solves the field on each one. Smaller pieces give a better answer and a slower solve.
- **Adaptive passes.** After each solve it finds where the error is largest, splits the pieces there, and solves again.
- **Convergence.** It stops when the answer stops changing between passes. The setting that controls this is **Delta S**: how much S11 may still move before it counts as settled.
- **Frequency sweep.** The mesh is built at one frequency, 2.45 GHz, then reused to work out S11 across a range of frequencies.
- **Far field.** From the field on the box, HFSS works out the beam at any distance: gain, beam width and pattern.

Model the air, not the copper

The wave lives in the air inside the horn and touches only the inside surface of the copper. So the model draws the air, and marks each wall as a perfectly conducting sheet (Perfect E). That is electrically what a copper horn is at 2.45 GHz, and it solves many times faster than meshing 0.5 mm metal.

3 The model

One horn, in its usual orientation: the wide side along x, the narrow side along y, the beam coming out along +z. The frame turns the horns a quarter turn, but that changes nothing about one horn on its own.



Side view, cut down the middle
(the y-z plane: the E-plane)

The top view is the same shape with
a = 86.4 at the pipe and
a1 = 263.8 at the mouth.

Red line: the wave port, where the
signal enters, under a metal cap.
Brass: the probe, on the centre line.

Variables

Type every size as a named variable, never as a bare number. Two of them, **plen** and **bshort**, are the ones you will sweep to tune the match, and a model built from bare numbers cannot be swept at all.

variable	value	what it is
freq	2.45 GHz	design frequency
a, b	86.4 mm, 43.2 mm	pipe, inside, wide and narrow
a1, b1	263.8 mm, 193.1 mm	mouth, inside
pe	91.5 mm	flare depth, pipe to mouth
Lg	115 mm	pipe length, back wall to the flare
bshort	43.7 mm	probe centre to the back wall. Swept
plen	28 mm	probe length into the pipe. Swept
drod	1.587 mm	probe diameter, 1/16 in brass rod
dpin	1.27 mm	SMA centre pin, inside the coax stub
ddiel	4.25 mm	Teflon diameter that makes the stub 50 Ω
clear	40 mm	air-box clearance round the horn

4 Building the model

1 Start the design

1. **Project** → **Insert HFSS Design**.
2. **HFSS** → **Solution Type** → **Modal** (listed as Driven Modal in older versions). Modal suits a single coax or waveguide feed.
3. **Modeler** → **Units** → **mm**, before drawing anything. Changing units later quietly reinterprets every number already typed.
4. **Project** → **Project Variables**, or type a name into any size box and HFSS offers to create it. Enter every variable in the table on the previous page.

2 Draw the air inside the horn

1. **Draw** → **Box** for the pipe: position $(-a/2, -b/2, -Lg)$, size (a, b, Lg) .
2. **Draw** → **Rectangle** at the throat: centred on the axis at $z = 0$, $a \times b$. Draw a second one at $z = pe$, $a1 \times b1$.
3. Select both rectangles, then **Modeler** → **Surface** → **Connect**. HFSS joins them into a solid: the flare.
4. Select the pipe and the flare, **Modeler** → **Boolean** → **Unite**, and rename the result **horn_air**. Material: vacuum.

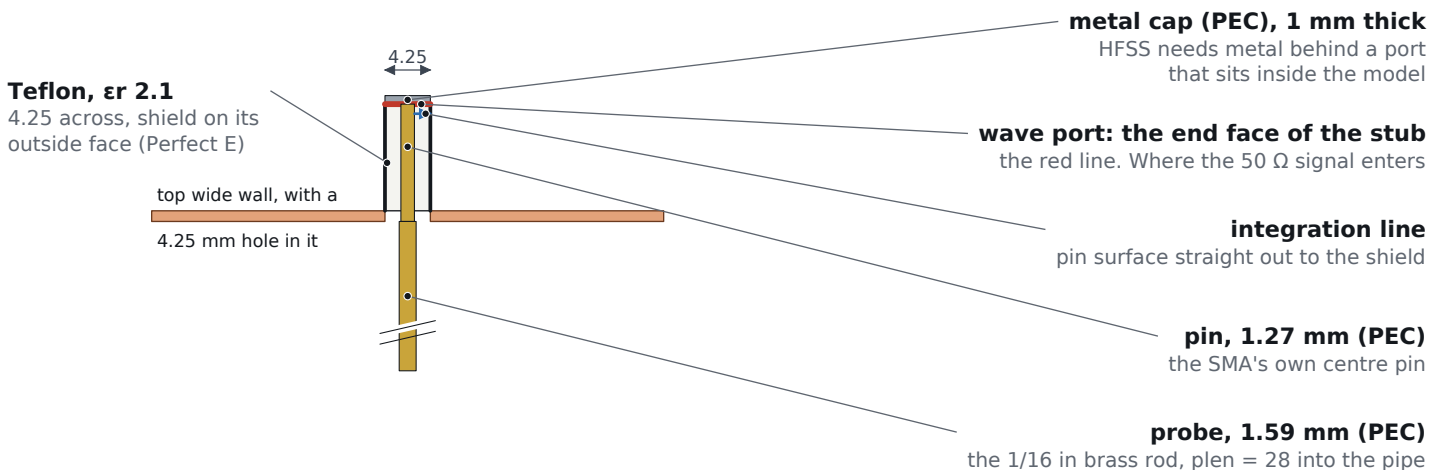
Check the flare: its sides should lean out 39.3° top and bottom and 44.1° at the sides. If Connect twists the solid, the two rectangles are not centred on the same axis.

3 Turn the walls into metal sheets

1. Select **horn_air** and switch to face selection (press **F**). Pick every outside face **except the mouth**: four flare faces, four pipe faces, and the back.
2. **Modeler** → **Surface** → **Create Object From Face**. These new sheets are the walls.
3. **Delete horn_air**. It was only a template; the air box fills that space later.
4. Select the wall sheets, right-click, **Assign Boundary** → **Perfect E**. Name it walls.

Later, for the real copper loss, change walls to **Finite Conductivity**, copper. Expect it to cost a few hundredths of a dB. Not worth it on the first run.

4 The feed: a short coax on the top wall



Cut through the top wall at the feed, drawn large. The stub stands for the SMA connector the build fits there.

1. **Teflon**. A cylinder along y , diameter $ddiel$, centred at $x = 0$, $z = -Lg + bshort$, from the top wall ($y = b/2$) up 10 mm. Material: Teflon (tm), $\epsilon_r 2.1$.
2. **Pin**. A cylinder, diameter $dpin$, on the same axis and the same length. Material: pec. Subtract the pin from the Teflon with **Clone tool objects** ticked, so the pin survives.
3. **Probe**. A cylinder, diameter $drod$, from $y = b/2$ down to $y = b/2 - plen$. Material: pec.

4. **Shield.** Assign Perfect E to the Teflon's curved outside face.
5. **Hole.** Cut a hole of diameter d_{diel} in the top wall sheet where the stub meets it. A probe that touches the wall gives a perfect-looking, completely false S11.
6. **Cap.** A pec disc, diameter d_{diel} , 1 mm thick, on top of the stub.

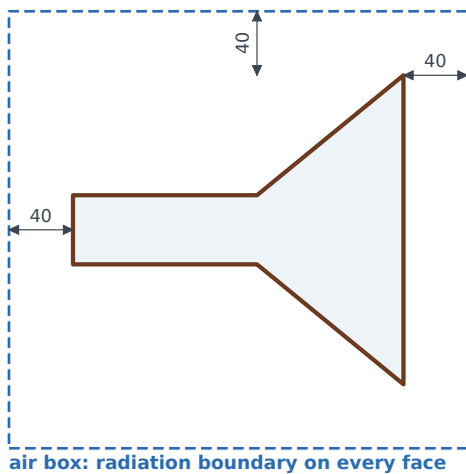
5 The wave port

1. Select the flat end face of the Teflon, just under the cap.
2. **Assign Excitation → Wave Port.** One mode. Draw the **integration line** from the pin's surface straight out to the shield. Keep **Renormalize to 50 Ω** on.

The port must cover the coax and nothing else

A port larger than the Teflon face stops behaving like a coax and the S11 means nothing. After the first solve, check **Results → Solution Data → Port Field Display**: mode 1 must look like spokes running from the pin to the shield.

6 The air box and the radiation boundary



How big the air box is

A radiation boundary must sit at least a quarter wavelength from anything that radiates: 30.6 mm at 2.45 GHz, and 34.1 mm at 2.2 GHz, the bottom of the sweep. 40 mm clears both.

Closer, and the boundary reflects some of the wave back. The pattern and S11 both move, and nothing warns you.

1. **Draw → Box** around everything, clear by the clear variable on every side. Material: vacuum.
2. Subtract the Teflon, pin and probe from the air box, with Clone tool objects ticked, so no two solids overlap.
3. Select the box's six outside faces, **Assign Boundary → Radiation**.

5 Solving it

7 Solution setup and frequency sweep

HFSS → Analysis Setup → Add Solution Setup, then add a frequency sweep to it.

setting	value	why
Solution frequency	2.45 GHz	the mesh is built here
Maximum number of passes	20	a ceiling, not a target
Maximum Delta S	0.02	0.01 for the final run
Minimum converged passes	2	one lucky pass is not convergence
Sweep type	Interpolating	few solves for a smooth curve
Sweep range	2.2 to 2.7 GHz, 501 points	see a mistuned dip, not just the band

8 Far-field setup

1. **HFSS → Radiation → Insert Far Field Setup → Infinite Sphere.**

2. A 3D sphere: θ 0 to 180°, ϕ 0 to 360°, 2° steps. Name it ff_3d.

3. The H-plane cut: $\phi = 0^\circ$, $\theta -180$ to 180°, 1° steps. Name it cut_H.

4. The E-plane cut: $\phi = 90^\circ$, same θ range. Name it cut_E.

9 Run it, then check the solver before believing it

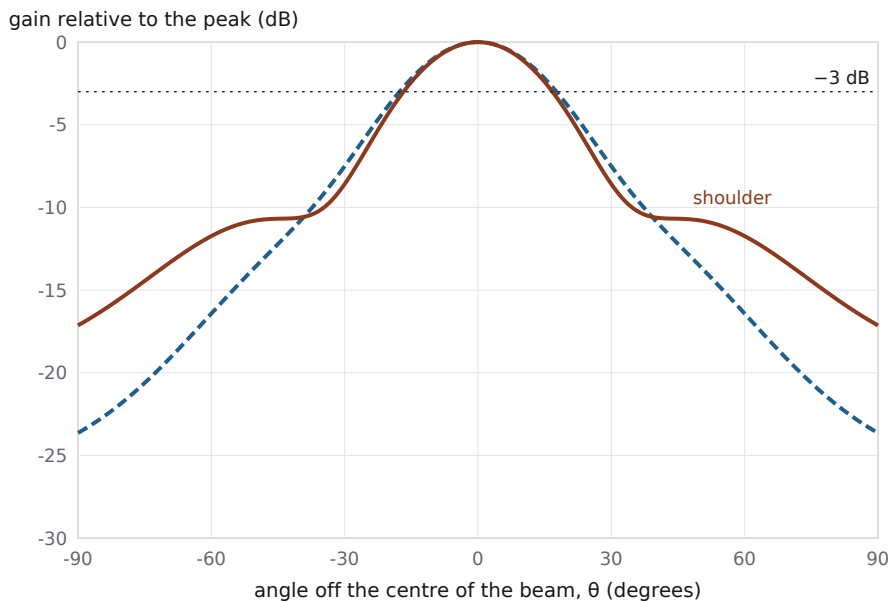
HFSS → Analyze All. Then work down this list in order.

	check	where	pass
1	convergence	Results → Solution Data → Convergence	Delta S under target, and the last passes barely changing
2	port mode	Solution Data → Port Field Display	spokes, pin to shield
3	mesh size	Solution Data → Profile	tens of thousands to a few hundred thousand pieces
4	gain vs the ceiling	Far field report, Realized Gain Total	below 16.3 dBi. Above it, the model is wrong
5	the numbers	the table on page 1	gain, both beam widths, S11

Reports: **Results → Create Modal Solution Data Report → Rectangular Plot** for S11 in dB, and **Results → Create Far Fields Report → Radiation Pattern** for Realized Gain Total on cut_E and cut_H. Use realized gain: it includes what the mismatch costs, which is the gain the radar actually gets.

6 What a good result looks like

HFSS should land close to these. The patterns below are computed by adding up the wave across the horn's mouth, the textbook method, independently of HFSS. They leave out what happens at the rim and behind the horn, so expect HFSS to differ beyond about 60° and to show a small back lobe.



What theory predicts

computed from the horn's mouth, not simulated

E-plane, solid line

beam 33.4° wide at -3 dB
shoulder near -11 dB, 40° to 55°

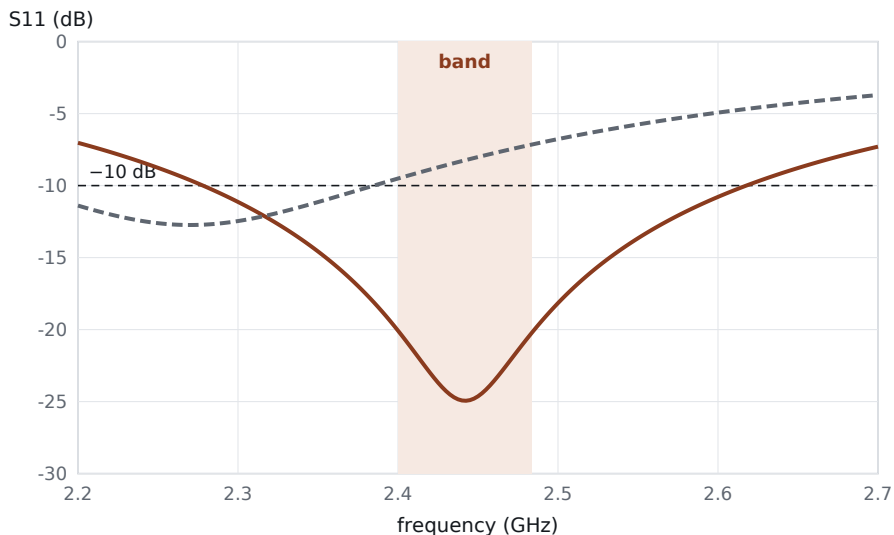
H-plane, dashed line

beam 35.2° wide at -3 dB
falls smoothly, no sidelobe

Peak gain 13.4 dBi

aperture efficiency 51%

The E-plane shoulder near -11 dB is real, not an error. It comes from the flare being short for its width, which is the trade that keeps this horn compact. HFSS may show it as a small bump or sidelobe at about the same level.



An illustration, not a result

the shape to look for

Solid line: a tuned probe

The dip sits inside the radar's band and the whole band is below -10 dB.

Dashed line: a mistuned probe

There is a dip, but at the wrong frequency. Plotting only the band would show a flat, poor line and no clue which way to move. Plot 2.2 to 2.7 GHz.

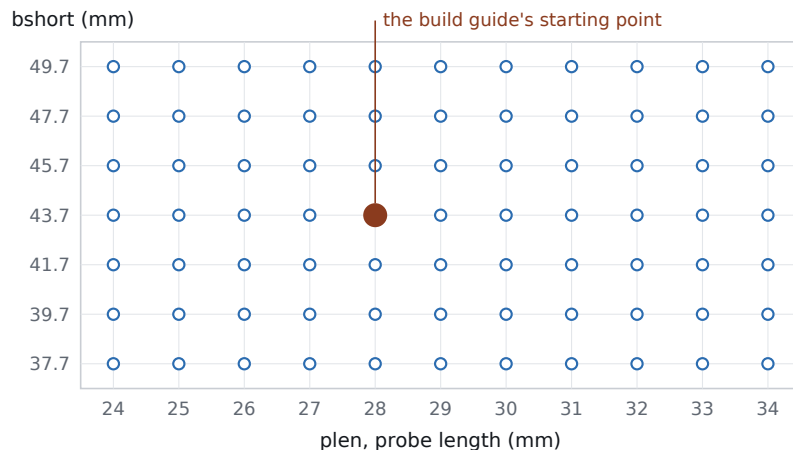
7 Tuning the match

Do not touch the flare: theory already sets it right. The match depends only on the feed, and there are exactly two knobs.

- **plen**, how far the probe reaches into the pipe, sets how strongly it couples to the wave. It mostly moves the bottom of the dip up and down.
- **bshort**, how far the probe is from the back wall, sets the timing between the wave going forward and the one bouncing off the back. It mostly slides the dip left and right.

10 Sweep both

Optimetrics → **Add** → **Parametric**, and add both variables as linear sweeps.



The tuning sweep

plen: 24 to 34 mm, 1 mm steps (11)
bshort: 37.7 to 49.7 mm, 2 mm steps (7)
= 77 solves, each with its own mesh.

Every circle is one full simulation.
Leave it running overnight.

At plen = 34 the probe tip still clears the far wall by 9.2 mm.

1. Plot S11 for every combination on one graph.
2. Keep the pair whose curve stays below -10 dB across the whole of 2.400–2.4835 GHz, with the dip near the middle.
3. If several pass, take the one with the most margin at the worst frequency, not the deepest dip.
4. Set those values as the defaults, re-solve at Delta S 0.01, and take the gain and patterns from that final run.
5. Give the chosen plen and bshort to the build: drill the feed hole at bshort and trim the rod to plen.

A deep dip is not the goal

-35 dB at one frequency and -8 dB at the band edge is worse than -14 dB everywhere. The radar sweeps the whole band, and the worst frequency is the one that counts.

8 The second model: transmit to receive

One question a single horn cannot answer: how much of the transmitted signal leaks straight into the receive horn. That leak is far stronger than the drone's echo, and the build needs it at **−35 dB or lower**.

11

Two horns, two ports

In the model: both horns in their usual orientation, mouths flush.



Why along x

On the frame every horn is turned a quarter turn, so their 263.8 side runs up and TX sits 290 mm above RX. Turned back to the model's orientation, that 'above' is the x direction.

Report S21, from port 2 into port 1, across 2.400–2.4835 GHz.

1. Save the tuned design as a copy. Select every part of the horn, **Edit → Duplicate → Along Line**, 290 mm along x.
2. Give the copy its own wave port. The original is port 1 (RX), the copy port 2 (TX).
3. Grow the air box so it still clears both horns by clear on every side.
4. Solve with the same setup and plot S21 in dB. Pass: −35 dB or lower across the band.

The build's foil sheet behind the horns and the frame are not in this model. Both should only improve the number, so a pass here is a safe pass.

9 Saving the results

Everything goes in **antenna/results/**. Right-click any report, **Export**, and save it as .csv beside a picture of it.

file	what
horn.aedt	the project, archived: File → Archive
s11.csv, s11.png	S11 of the tuned horn, 2.2 to 2.7 GHz
gain-eplane.csv, gain-hplane.csv	realized gain on cut_E and cut_H at 2.45 GHz
patterns.png	both cuts on one plot
probe-sweep.csv	the parametric sweep: S11 for every plen and bshort
isolation.csv	S21 between the two horns

Results record

Fill this in from the final, tuned run. It is the page the LaTeX report is built from.

result	target	HFSS
realized gain at 2.400 GHz	13.4 ± 1.5 dBi	
realized gain at 2.442 GHz	13.4 ± 1.5 dBi	
realized gain at 2.483 GHz	13.4 ± 1.5 dBi	
beam width, E-plane	$33.4 \pm 4^\circ$	
beam width, H-plane	$35.2 \pm 4^\circ$	
E-plane shoulder or first sidelobe	about -11 dB	
worst S11 in 2.400–2.4835 GHz	-10 dB or lower	
chosen plen	start 28 mm	
chosen bshort	start 43.7 mm	
TX to RX, worst S21 in band	-35 dB or lower	
final passes, final Delta S	converged, ≤ 0.01	

Common mistakes

symptom	likely cause
gain above 16.3 dBi	a wall missing its Perfect E, or the air box too small and reflecting
S11 near 0 dB everywhere	the probe touches the wall, or the port covers more than the Teflon face
a perfect-looking dip, far too deep	the probe is shorted to the wall: check the hole
the H-plane beam comes out narrower	the cuts are swapped, or the pipe was drawn a quarter turn round
results change when the air box grows	it was too close. Keep clear above 34 mm
millions of mesh pieces	a thin solid somewhere: walls must be sheets

Words used in this guide

word	means
dB, dBi	a ratio on a log scale. -10 dB is a tenth, -20 dB a hundredth. dBi is gain compared with an antenna that radiates evenly in every direction
S11	the fraction of power that bounces back from the antenna into the cable
S21	the fraction that leaks from one antenna's port into another's
realized gain	gain after taking off what the mismatch reflects
waveguide	the metal pipe at the back of the horn
Perfect E	a surface HFSS treats as perfect metal
wave port	where the signal enters the model
radiation boundary	a surface that lets waves leave the model without reflecting
mesh, pass	the small pieces space is cut into, and one round of solving them
Delta S	how much S11 changed between the last two passes
far field	the beam as seen from a long way off